

Executive Business Report
End-to-End Sales Forecasting & Demand Intelligence System
Prepared For: Head of Supply Chain & Chief Financial Officer (CFO)
Prepared By: Sujit Kushwaha

1. Executive Summary

This project was carried out to help the business make better inventory and sales planning decisions using historical sales data. We analyzed four years of retail sales, identified demand patterns, built forecasting models to predict the next three months of sales, detected unusual sales events, and grouped products based on their demand behavior. An interactive dashboard was also developed to help business users monitor sales performance and forecasts without requiring technical knowledge. The results provide practical insights for inventory management, demand planning, and operational decision-making.

2. Key Findings from Exploratory Data Analysis (EDA)

The sales analysis revealed several important business insights:

- Sales showed an overall upward trend over the four-year period, indicating consistent business growth.
- Technology products generated the highest revenue among all product categories.
- Sales increased significantly during seasonal and festive periods, showing clear demand seasonality.
- The West region consistently contributed the highest sales compared to other regions.
- Average shipping time remained stable across most regions, indicating consistent logistics performance.
- Monthly sales fluctuations suggest that inventory planning should consider seasonal demand patterns.

3. Three-Month Sales Forecast

Three forecasting models (SARIMA, Facebook Prophet, and XGBoost) were developed and evaluated using MAE, RMSE, and MAPE. Based on the evaluation results, **[Best Model Name]** achieved the lowest forecasting error and was selected as the production model.

Forecast Summary

Month Forecast Sales 95% Confidence Range

Month 1 **[Forecast 1]** **[Lower 1] – [Upper 1]**

Month 2 **[Forecast 2]** **[Lower 2] – [Upper 2]**

Month 3 **[Forecast 3]** **[Lower 3] – [Upper 3]**

Plain Language Interpretation

The forecast indicates that sales are expected to remain **stable/grow** over the next three months. Even if actual sales vary slightly, they are expected to remain within the confidence ranges shown above. This forecast can be used to improve purchasing, inventory planning, and staffing decisions while reducing the risk of stock shortages or excess inventory.

4. Top Three Sales Anomalies

The anomaly detection models (Isolation Forest and Z-Score) identified several unusual sales periods.

Anomaly 1

- **Period:** [Month/Date]
- **Observation:** Sales were significantly higher than normal.
- **Likely Cause:** Seasonal promotion, festival sales, or special discount campaign.

Anomaly 2

- **Period:** [Month/Date]
- **Observation:** Sales dropped sharply below expected levels.
- **Likely Cause:** Supply chain disruption, stock shortage, or reduced customer demand.

Anomaly 3

- **Period:** [Month/Date]
- **Observation:** Sudden increase in order volume.
- **Likely Cause:** Bulk corporate purchase or successful marketing campaign.

These events should be monitored regularly because they can significantly affect forecasting accuracy and inventory planning.

5. Product Demand Segmentation

Products were grouped into demand segments using K-Means Clustering based on:

- Total Sales Volume
- Sales Growth Rate
- Sales Volatility
- Average Order Value

Demand Segments

Demand Segment	Business Meaning	Recommended Stocking Strategy
High Volume, Stable Demand	Consistent best-selling products	Maintain high inventory with regular replenishment.
Low Volume, High Volatility	Unpredictable demand	Keep limited stock and monitor frequently.
Growing Demand	Increasing customer interest	Increase inventory gradually to avoid stock-outs.
Declining Demand	Decreasing sales trend	Reduce inventory and avoid overstocking.

This segmentation helps optimize warehouse space and reduce inventory holding costs.

6. Business Recommendations

Recommendation 1: Increase Inventory for High-Demand Products

The analysis shows that high-volume product segments consistently contribute the largest share of revenue. Maintaining higher inventory levels for these products will reduce stock-out risk and improve customer satisfaction.

Recommendation 2: Use Forecasts for Inventory Planning

The selected forecasting model achieved the lowest prediction error (**RMSE = [Your RMSE]**), making it suitable for short-term demand planning. Procurement and warehouse teams should use these forecasts to optimize monthly inventory levels.

Recommendation 3: Monitor Sales Anomalies

Anomaly detection identified unusual sales spikes and drops that may indicate promotions, supply issues, or operational problems. Regular monitoring of these events can improve forecast accuracy and support faster business decisions.

7. Risk / Limitation

This forecasting system relies entirely on historical sales data. Unexpected events such as economic changes, new competitors, policy changes, supply chain disruptions, or major promotional campaigns may cause future sales to differ from the predicted values. Therefore, forecasts should be reviewed regularly and updated whenever new data becomes available.

8. Conclusion

The End-to-End Sales Forecasting and Demand Intelligence System provides a practical solution for retail demand planning. By combining sales forecasting, anomaly detection, and product demand segmentation, the business can improve inventory management, reduce unnecessary stock, and make data-driven decisions with greater confidence. The interactive dashboard further enables managers to monitor performance and forecasts without requiring technical expertise, making it a valuable decision-support tool for both Supply Chain and Finance teams.